

GEOG 531 Quantitative Methods In Geographic Research

Statistical Data Types and Nature of Spatial Data

Learning check

Intro to Modern Statistics, Chapter 1, Q 12.

- a. What does each row of the data frame represent?
- b. How many participants were included in the survey?
- c. Indicate whether each variable is numerical or categorical. If numerical, identify as continuous or discrete. If categorical, indicate if the variable is ordinal.

Learning check

Intro to Modern Statistics, Chapter 1, Q 14.

- a. List the variables used in creating this visualization.
- b. Indicate whether each variable is numerical or categorical. If numerical, identify as continuous or discrete. If categorical, indicate if the variable is ordinal.

Data and Dataset

- Data: Measurable information
 - e.g., population of a city
- Dataset: A collection of data consisting of observations, variables and data values

Home sales of Milwaukee, Wisconsin, 2014 (subset)

Address	Sqft	Sale \$
129 W BOLIVAR AV	1,246	130,000
1336 W VAN NORMAN	1,143	87,000
137 W UNCAS AV	1,521	150,000
1409 W WANDA AV	1,059	168,000
1416 W HOWARD AV	1,958	122,500
1420 W FOSTER AV	1,208	136,000

- **Observations**: the elements under the study
- **Variables**: the properties of observation that measured
- **Data values**: the measurement of the properties of the observations

Statistical Data Types

- Nominal, Ordinal, Interval, and Ratio
 - Indicate different levels of measurement/information
- Discrete and Continuous
 - Discrete data: can only take certain values(finite number of values), e.g., zip codes, number of sunny days in a year, COVID cases
 - Continuous data: can take any value(within a range, infinite number of values) e.g., height, temperature
- Quantitative and Qualitative
 - Quantitative: deals with numbers
 - Qualitative: deals with descriptions, e.g., color of eyes

Typology of variables: Intro to Modern Stats, Ch 1

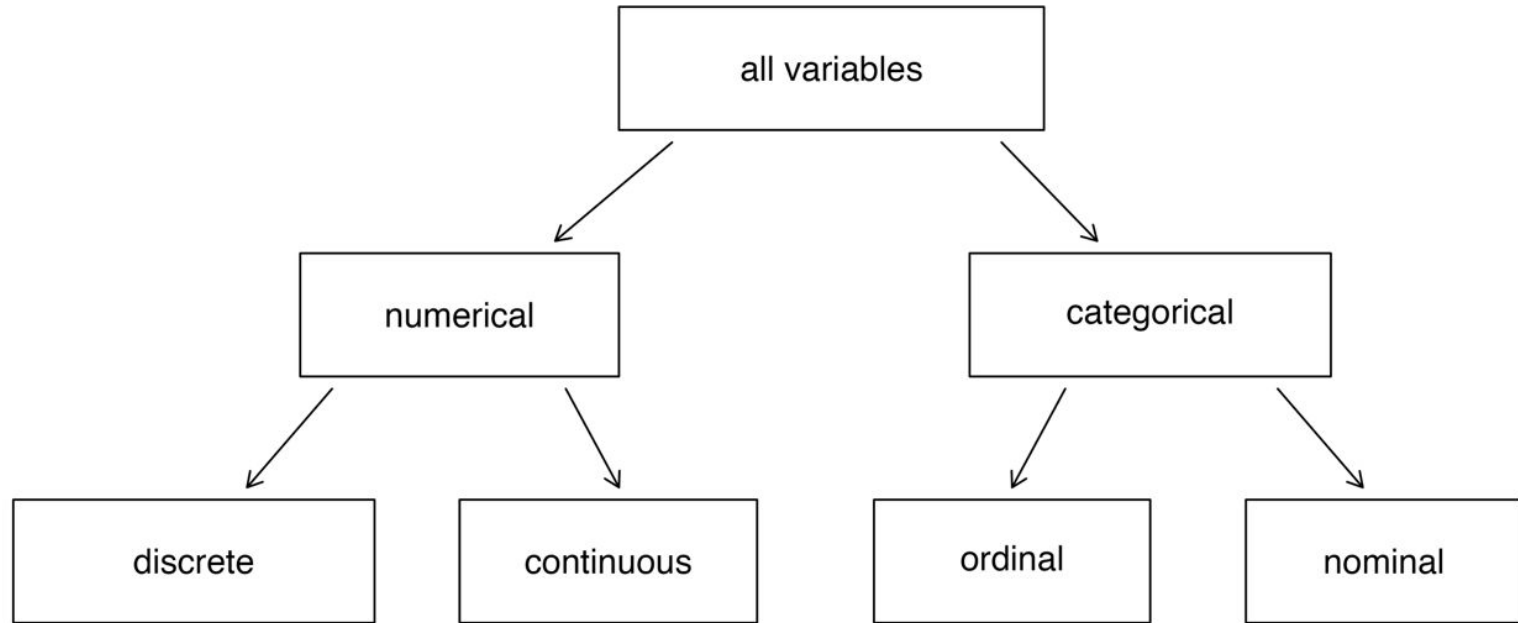


Figure 1.1: Breakdown of variables into their respective types.

Nominal Data

- Also known as Categorical Data
- Observations are placed into a set of unordered categories
 - mutually exclusive
 - e.g., gender: {male, female},
geographic location: {North, South, West, East}
 - Qualitative or quantitative? → Qualitative
- Binary Data: when only two categories:, e.g., yes/no, success/failure

Ordinal Data

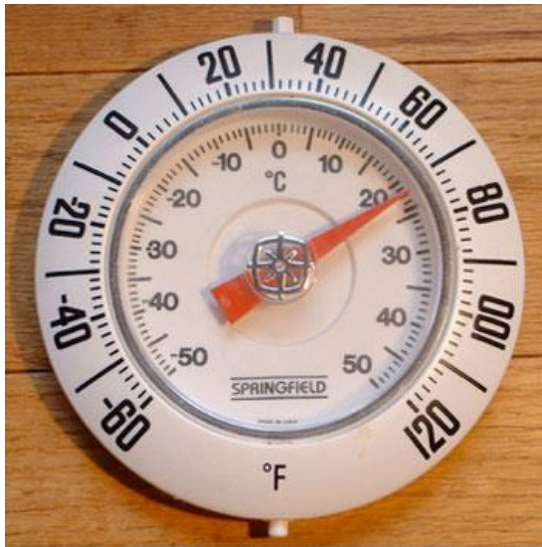
- Observations are ordered or ranked
 - ordered by groups: weak order (e.g., low, average, high)
 - ordered by individual ranks: strong order (e.g., 1st, 2nd, 3rd, ... last)
 - Comparable: e.g. A is larger than B. But cannot tell how by how much.
 - Qualitative or quantitative? → Qualitative

Country ranked by Area:

Rank	Country	Area(sq km)
1	Russia	17,075,200
2	Canada	9,984,670
3	United States	9,826,630
4	China	9,596,960
5	Brazil	8,511,965

Interval Data

- Not only indicates order, but also the difference between observations
- No natural zero point
- Can add/subtract, **cannot** multiply/divide



Example: temperature in Celsius or Fahrenheit

0 °C doesn't mean "no temperature"

30 °C is 15 °C warmer than 15 °C , but is not twice as warm as 15 °C (15 °C vs -15 °C)

Quantitative or Qualitative? → Quantitative

Ratio Data

- Interval data with a natural zero point
- $+$, $-$, $\times$, $\div$ is meaningful
- Most numerical data is ratio data
 - population in an area
 - height, weight ...
 - temperature in Kelvin: absolute zero(no energy)

Quantitative or Qualitative? → Quantitative

Data Types Summary

	Level of information			
	Low			High
Properties	Nominal	Ordinal	Interval	Ratio
Distinctness ($= \neq$)	✓	✓	✓	✓
Order ($< >$)		✓	✓	✓
Addition ($+ -$)			✓	✓
Multiplication ($\times \div$)				✓
	qualitative discrete		quantitative continuous/discrete	

Data Type Exercise

For the following measurements, identify their data types:

- nominal, ordinal, interval or ratio?
- discrete or continuous?
- qualitative or quantitative?

- **Zip code**
- **Bronze, Silver, and Gold medals as awarded at the Olympics**
- **Temperature in Fahrenheit**
- **Number of patients in a hospital**
- **Angles as measured in degrees between 0° and 360°**

Data Type Exercise

For the following measurements, identify their data types:

- nominal, ordinal, interval or ratio?
- discrete or continuous?
- qualitative or quantitative?

- **Zip code**

- nominal, discrete, qualitative

- **Bronze, Silver, and Gold medals as awarded at the Olympics**

- ordinal, discrete, qualitative

- **Temperature in Fahrenheit**

- interval, continuous, quantitative

- **Number of patients in a hospital**

- ratio, discrete, quantitative

- **Angles as measured in degrees between 0° and 360°**

- ratio/interval , continuous, quantitative